

Interactive Handwriting Device for Enhancing Active Recollection of Character Forms by Voice Assistance for Chinese Character Learning

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ABSTRACT

Learning the Chinese characters used in Japan and China is a relatively difficult task, and even native speakers often have difficulty recalling them. Despite the existence of numerous platforms for learning Chinese characters, there is still a lack of interactive resources that can help with retrieving information during composition and improving retention simultaneously. In this study, we propose an interactive handwriting-support device that provides verbal hints about Chinese characters' compositions and how to write the character. The device has the shape of a pen and can audibly indicate to the user how to write a character while writing. From the experiment, it is found that retention tended to be highest when the character composition was guided only by a voice assistance, compared to when the correct character form was presented visually. This result suggests that presenting human-like cues through speech may promote active recollection activities compared to presenting visual information.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Human computer interaction (HCI); Interaction paradigms; Natural language interfaces.

KEYWORDS

Interaction device, Language learning, Chinese character, Character amnesia

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1 INTRODUCTION

Chinese characters are the second-most used natural language writing system after the Latin alphabet, with approximately 1.34 billion users worldwide. However, Chinese characters are ideographic, and their large number makes them difficult to master. According to

the Foreign Service Institute (FSI), an agency of the United States Department of State [1], Chinese and Japanese, which regularly use Chinese characters, are included in Category IV, the most difficult languages for English speakers to master. In addition, it is known that even native users of Chinese characters are becoming less proficient in their use. Chinese and Japanese writing literacy is taught by rote memorization, and schoolchildren learn how to write characters by repeating them by hand [2]. However, as people have become less dependent on handwriting and more willing to use computer input methods, they are no longer exposed to the reinforcements necessary to maintain their ability to write characters. Hence, they have forgotten the forms of Chinese characters, and although they can visually recognize characters, without the assistance of information devices, they are no longer able to hand-write characters that are usually used infrequently. This phenomenon is called character amnesia [3], and it has been attracting attention as a new issue caused by the development of information technology in the East Asian cultural sphere [4]. Therefore, it would be useful for both non-native learners and native speakers of Chinese characters if they could receive assistance in writing characters in situations where they cannot recall their handwritten forms.

Based on the background, various systems have been proposed to support the learning of Chinese characters by providing visual information about their glyphs [5-10]. While visual presentation approaches are useful for retrieving information to look up the shape of unfamiliar Chinese characters, they are not sufficient to promote long-term memory retention because many methods lack a mechanism to encourage the effort of recalling the correct form of the characters. In addition, there are proposals to promote interest in and retention of Chinese characters, for example, visual presentation of the origin of Chinese characters [10], use of songs expressing Chinese characters [11], and mechanism to force the recall of correct character form during work [12]. However, these proposals are heavy burdens to be used for information retrieval in daily writing. Therefore, it remains a challenge to construct a system that supports both simple information retrieval and retention in order to effectively learn and recall Chinese characters.

To address these problems, the use of conversational agents which respond interactively to user queries could be a promising solution. The development of speech recognition and natural language processing technologies has led to the popularity of interactive information retrieval applications, such as intelligent speakers (e.g., Amazon Alexa). Interactive information retrieval provides users with information in a more anthropomorphic and understandable format, similar to receiving information from other people [13-15]. However, these voice-based interactive user interfaces may also have the potential to improve memory retention. In psychological

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research, the modality effect is well known [16], and it suggests that text presented in an auditory format is more likely to be remembered than text presented visually. In addition, associating words with visual and auditory features is thought an effective method for learning foreign languages [17]. Therefore, an auditory approach to learning support may be promising. Furthermore, because information is more abstracted in auditory communication than in visual communication in many cases, users may be encouraged to think independently. Indeed, studies of learning support using interactive agents have confirmed that the modality effect improves memory retention [18, 19].

In interpersonal interactions in the Chinese character community, there are often situations in which people are taught how to write Chinese characters based on explanations using the structural relationships of character forms, such as, “The Chinese character for the organ called lung (肺) is represented by the characters for moon (月) and city (市)”. Thus, especially for native users of Chinese characters, the structure of the character form often serves as a cue for remembering the character [20–22]. This is supported by the fact that structure-based input methods (e.g., Cangjie input method) are used for Chinese characters in information devices. Awareness of such combinations of characters is considered important in the acquisition of complex Chinese characters [22] and it is expected that explanations that raise awareness of the structure of a Chinese character to someone who wants to learn will aid in recall and retrieval.

Based on this idea, we propose an interactive handwriting-support device that can provide hints of how to write Chinese characters through voice assistance, just as a human explains through speech (Figure 1A). The device is in the form of a ballpoint pen, and it can acquire information about how to write Chinese characters through speech interactions in the form of an intelligent speaker. Because the device is in the form of a writing instrument, it can be used as a reference while handwriting, and it is portable. When searching for the character form using other terminals, such as smartphones, it is necessary to put down the writing utensil and focus attention on the device to search. Conversely, this device allows users to search and work simultaneously. In this paper, we investigate the effects of visual and auditory information presentation and retention using the prototype system. Our goal of this research is to clarify the usefulness of guidance based on human-like “speech” from the writing instrument as a method to encourage users to remember characters they cannot recall.

A secondary goal of this research is to explore a model of applications that supports users’ independent retrieval activities, considering modality effects. In general, visual information that can be reviewed is used in information presentation devices for healthy people that are intended to convey information accurately. However, a presentation strategy that uses auditory information in situations that should be presented visually may provide a means to encourage user-initiated behavior to acquire missing information naturally. Therefore, studying the differences in recall and effort between modalities in realistic information retrieval situations is expected to provide useful knowledge for designing interfaces that encourage participation while considering modality effects.

2 SYSTEM DESIGN AND IMPLEMENTATION

In this paper, we propose a device that supports users’ recall of character forms when handwriting. When the user speaks to the pen, the device guides the user with information about the composition of character forms by voice assistance. At first glance, information presented through speech is difficult to understand as a guide from an information system, because it does not directly present the correct answer. However, in the proposed concept, we intentionally adopt such an interaction style, as it is a natural style of information transfer through interpersonal interaction that can elicit users’ independent efforts to recall the content of Chinese characters.

To investigate the validity of the above concept, we implemented an experimental prototype device. This device is a ballpoint pen and can be used as a writing instrument. As shown in Figure 1A, the device consists of a processor, speaker, microphone (M5Atom Echo, M5stack Inc.), and LCD screen with wireless communication. As the purpose of the verification in this paper is to investigate the main effects and interactions between visual and auditory presentation in Chinese character recall, the prototype hardware is equipped with an LCD screen and an acoustic playback device inside the pen to support both types of information presentation. In the experiments described below, LCD screen is used for responding to the user’s call and provide information in a visual way (Figure 1 C and D).

In this section, we describe the configuration of the experimental software used in the evaluation experiments presented in this paper. The evaluation experiments described in this paper were conducted using the Wizard-of-OZ method, in which the experimenter manually manipulates the behavior of the system from a remote location. There are two types of experimental software: firmware, which runs inside the device, and software, which runs on a remote PC and allows the experimenter to issue instructions to control the device. The firmware running inside the device consists of software that controls the speaker as a Bluetooth child device and a web server that accepts instructions from an external remote-control terminal and operates the LCD screen display. This firmware allows the device to act as a Bluetooth speaker, and audio played on the remote experimenter’s PC can be played in real-time on the device. In addition, the web server running inside the device has a function to display images and text prepared in advance in response to HTTP requests sent from the remote-control terminal. Then, by issuing a request from the remote-control PC, the screen displays contents, and the speech timing can be remotely controlled.

3 USER STUDY

3.1 Purpose and Hypothesis

We conducted an experiment to clarify the usefulness of information guidance based on human-like “speech” from the writing instrument as a method to encourage users to remember Chinese characters they cannot recall. As mentioned at the introduction, the character amnesia is a prominent problem among native Chinese character users, such as Japanese. Therefore, in this experiment, we investigate the usefulness of the proposed method as a support device for the problem of character amnesia, especially among native Japanese users. Therefore, 18 Japanese university students (14 males and 4 females) with normal vision and hearing from the

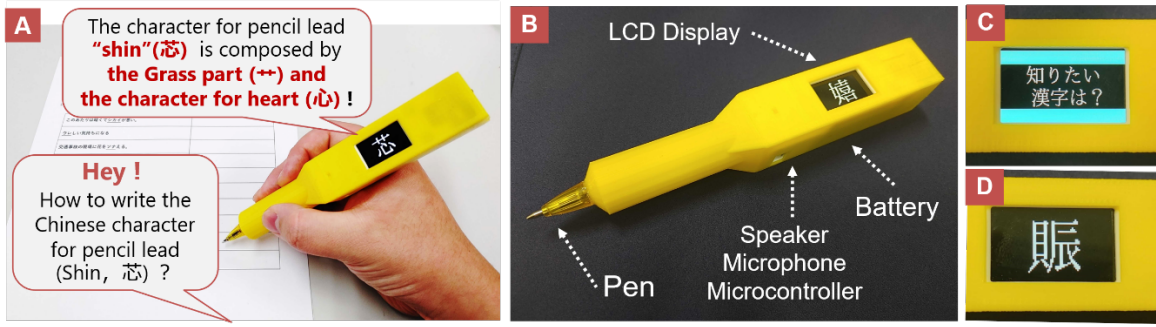


Figure 1: (A) Example of use of the proposed device, (B) Configuration of the proposed system, (C) Command waiting screen/The screen shows the Japanese sentence, “What character do you want to know?” (D) An example of character presentation via screen.

author’s university participated in the experiment, wherein we test the following hypotheses:

- H1: Voice-assisted only instructions on the proposed device are perceived as more difficult to understand than visual instructions, but provide a relatively high level of understandability.
- H2: Voice-assisted-only instructions on the proposed device provide a sense of assistance, but to a lesser degree than visual instructions.
- H3: Voice assistance from the proposed device provide the user with a sense of familiarity and support.
- H4: When instructions are given only through voice assistance from the device, users actively recall information about the character form and retention is enhanced.

This experiment’s hypotheses are focusing on the possibility that guiding via human-like speech creates a different value than visual presentation such as explored in many existing studies. Guidance based on visual information can strictly inform the correct character form. In other words, this guidance method provides the answer directly. Therefore, it is considered that user can feel the visual guidance as convenient method to recall unfamiliar Chinese characters. On the other hand, voice-based guidance does not strictly provide information on the external form of characters, however, this communication method is often used to receive information on unfamiliar Chinese characters from others, as mentioned above. Therefore, human-like voice guidance is also considered to be relatively understandable (Hypotheses H1). Similarly, as voice guidance provides hints for approaching the correct answer, it is considered that the user may experience a sense of assistance, although this is inferior to visual presentation, which provides the correct answer directly (Hypotheses H2).

The voice-based guidance method on the proposed system is closer to interpersonal communication than visual guidance. Therefore, it is believed that it will form an anthropomorphic representation of the device to the user, similar to that of existing anthropomorphic conversational agents [13, 15, 23, 24], and this may create a sense of familiarity and support (Hypotheses H3). Furthermore, because voice-based guidance does not directly convey the character form to the user, the user must recall the character shapes when receiving the information guidance. Therefore, it is possible that

the user’s effort to recall the content of the Chinese characters will enhance retention of the content when the information is provided via voice assistance (Hypotheses H4).

3.2 Preliminary Survey

In this experiment, the above hypotheses were tested by having the participants perform a Chinese character-writing task using the proposed device and analyzing their impressions and performance during the task. The experiment was designed as a two-stage experiment with a pre-survey and a main survey to set up a situation in which the participants recall the Chinese characters. The main survey also includes a post-survey to observe changes in performance from the pre-survey.

3.2.1 Pre-survey to determine participants’ knowledge of Chinese characters. In the pre-survey, participants were asked to answer 50 questions about translating words written in Japanese katakana into Chinese characters to determine their knowledge of the characters. The questions in the pre-survey were designed by selecting questions from the most famous and official Japanese Chinese Character Proficiency Test (Japan Kanji Aptitude Test), which is a series of questions on translating katakana into Chinese characters, according to level (Level 8, 5, 3, to Pre-1). Table 1 shows the question levels and the percentage of correct answers given to the participants in the experiment.

3.2.2 Design of the questions for the main experiment. Based on the results of the dictation questions in the pre-survey, 24 questions, mainly those with a low percentage of correct answers by the participants, were used as the basis for the dictation questions in this survey. To reproduce the use case of “asking the system how to write the character when the user cannot recall it,” the questions were designed to include questions that the user could answer by themselves. Specifically, six low-difficulty questions with a correct response rate of 80% or more and six high-difficulty questions with a correct response rate of 10% or less were selected based on the average correct response rate in the preliminary survey. In addition, the other six questions with relatively high difficulty questions with a correct response rate of 50% or less were added. Moreover, by asking users to write Chinese characters in the pre-survey, it is feared that they will become interested in the Chinese characters

Table 1: Breakdown of the difficulty level of the 50 Chinese character questions used in the pre-survey and post-survey of the user study, the number of questions, and the percentage of correct answers by the participants in the pre-survey.

Level	Difficulty of these questions in Japan	Num of questions in preliminary survey	percentage of correct answers in all participants (%)
8	Learned up to the third grade of elementary school (age 9)	5	93.3
5	Learned up to the sixth grade of elementary school (age 12)	5	81.1
3	Secondary school level	10	57.8
Pre-2	High school level	10	51.1
2	High school graduate / university / general public level	10	46.7
Pre-1	University/general public level	10	2.8

and learn how to write them by looking them up on their own. Therefore, three questions from Level 5 and three questions from Level Pre-1 of the Chinese Character Proficiency Test were added to the questions in this survey as new questions that users will encounter for the first time. Finally, 30 questions were prepared for this survey. The questions used in the main survey and the percentage of correct answers in the pre-survey are presented in the Supplementary material.

The participants were asked to partake in the main experiment about one to two weeks after the preliminary experiment. In the main experiment, the participants were asked to perform a Chinese character-writing task for the main experiment using the proposed system. In addition, after experiencing the system, participants were asked to complete the same 50 Chinese character writing task from the pre-survey as a post-survey. The details of the experiment are described in the next section.

3.3 Main experiment

3.3.1 Experimental conditions and settings. To test hypotheses H1 to H4, we conducted an experiment in which users were asked to respond to a Chinese character-writing task using our proposed device. Then, we investigated their impressions of the information provided by the device and the change in user performance from the pre-survey to the post-survey via comparison. To investigate each hypothesis, the following three experimental conditions were set up and compared in terms of the method used to guide users in writing Chinese characters.

Voice Condition: When a user asks the system how to write Chinese characters, the system guides the user with information about the Chinese characters by presenting only voice assistance.

Visual Condition: When the user asks the system how to write Chinese characters, the system guides the user by presenting the correct answer image on the device’s LCD screen, along with an electronic sound notification.

Combined Condition: When the user asks the system how to write Chinese characters, the system guides the user with information about the Chinese characters by both voice assistance and visual feedback on the LCD screen.

Participants were asked to face each of the above three guidance patterns by performing three Chinese character-writing tasks using the proposed pen device. The order in which participants experienced the three guidance patterns was determined differently for each participant. For the Chinese character dictation task,

the 30 questions described in section 3.2.2 were divided into three sets, so the distribution of question levels included was even, and 10 questions were created for each of the conditions of the guidance method. In particular, the question sets for each condition are designed to ensure that they always include at least two high-difficulty questions with a correct response rate of at least 80% and two low-difficulty questions with a correct response rate of 10% or less in the pre-survey. The questions in this experiment and the corresponding conditions for each question are given in the Supplementary material of this paper. The experiment was conducted using the Wizard-of-Oz method, with the experimenter remotely controlling the device from a location out of the participants’ sight.

3.3.2 Experimental system settings. To make the proposed system appear to run autonomously for the participants, instructions for using the system to the participants and manual manipulation procedures by the experimenter are designed. In this experiment, participants were instructed to say “Hey” to the system when it asked for Chinese characters they did not understand, regardless of the conditions. This is equivalent to the wake-word of a smart speaker. At this point, the experimenter manipulated the device’s LCD screen to switch to a special standby screen waiting for the user’s question (Figure 1C). When the user asks a question, the device plays a preset audio/visual response that matches the content of the question on the device’s speaker and LCD screen (Figure 1D). For each of the 30 questions used in this experiment, 30 patterns of synthetic speech files in total were created to explain how to write Chinese characters. These audio files were used to present information in both the voice and combined conditions. By examining the content of speech during voice assistance, all guidance scripts were designed to include explanations based on the radicals of Chinese characters (e.g., “The character for lung (肺) is written City (市) to the right of the character for Moon (月)”). Figure 1A shows examples of utterances used to explain how Chinese characters are written. In the visual condition and the combined condition, the image of the correct Chinese character is presented on the device’s LCD screen. In the visual condition, the device is set to play an electronic sound using a chime instead of a synthetic voice.

3.3.3 Evaluation metrics. After each trial, participants were asked to complete a questionnaire concerning their impressions of the device. The list of questions is shown on Table 2 along with their results. Each question is designed to address the specific hypothesis. Specifically, Q1 on comprehension (H1), Q2 on sense of support

(H2), Q3 on familiarity (H3) and Q4-5 on effort during recall (H4) are asked. The questions are evaluated on a 7-point Likert scale (7 being the most positive). In addition, 50 Chinese character dictation questions in the pre-survey were asked again as the post-survey, and the change in the number of correct answers was used as the target of evaluation to investigating hypothesis H4.

3.3.4 Procedure. Participants were asked to enter a small room equipped with a surveillance camera, and they were then asked to use the proposed handwriting device to answer the questions. After handing over the question paper, the experimenter told the participant that the experimenter would leave the room so they could concentrate, moving to a control console outside the small room. The experimenter constantly monitored the participant's work via a camera image to determine the timing of manual operation of the system. When the participant requested access to the device using voice commands, the experimenter manually operated the device using remote control software. After confirming that the participant had completed the question paper, the experimenter asked the participant to answer the questionnaire, thus completing the trial. This series of trials was conducted three times, varying the pattern of information delivery to the device. The experiment ended with a post-survey in which participants were asked to complete the Chinese character-writing task given to them in the pre-survey.

4 EXPERIMENTAL RESULT

4.1 Users' impression from the questionnaire-based survey

Table 2 shows the results of the questionnaire. A Friedman test was performed on the mean score of each question, and significant difference in Q1-Q4 and significantly trending in Q5 are observed respectively. In Q1, the mean scores were higher in the order of combined, visual, and voice conditions. In addition, the difference between the voice and combined conditions was significant. However, the mean score was higher than 6 in all conditions. Therefore, comprehension of the presented content was highly rated, even in the audio condition, which did not provide the information about the character's shape strictly. A similar trend was found for Q2, which asked about the feeling of having received support, and as in Q1, the main effect was significant. However, the significance of the difference could not be determined from the results of the multiple comparisons by using Holm method. In terms of mean score, the score for the voice condition was lower than those for the other two visual presentation conditions, but it was still relatively high (5.83). These results suggest that presenting information by talking to the participants may give them a sense of support. However, it reduced their comprehension and sense of support compared to the other conditions that provide "correct" answers. Thus, hypothesis H1 and H2 were supported.

Q3, an item concerning familiarity with the system, showed a tendency for scores in the combined and voice conditions to be higher than those in the visual presentation condition, where the difference in scores between the combined and visual presentation conditions was significant. This result suggests that presenting information with voice assistance about the character from enhances familiarity with the system. Thus, hypothesis H3 was supported.

Q4 and Q5 are items concerning the subjectivity of the experimental participants during the work. In the result of both question, participants tended to make more effort to recall Chinese characters in the voice condition than in the other presentation conditions, a result consistent with hypothesis H4.

4.2 Changes in participant's performance

We quantitatively examined the effects of the system intervention by looking at the number of correct answers in the Chinese character writing tasks in the pre-survey and post-survey. In this process, the 24 questions (8 questions in each condition) duplicated in the main survey are selected as sample. Then, we checked whether there was a difference in the average number of correct answers, depending on the presentation method between pre- and post-result in the selected questions. Table 3 shows the change in the percentage of correct answers to the Chinese characters included in each condition in this study. Then, a two-way ANOVA based on pre/post factors (two levels) and presentation method (three levels) was conducted concerning the average number of correct answers for each condition. The results showed a main effect for the pre- and post-response factors ($p < .001$), and the interaction was significant ($p = .04$). This result suggests that the average number of correct answers in the post-survey are significantly higher than the pre-survey, regardless of the condition, but the effect varied by presentation condition. Additionally, a simple effect test was performed to examine the interaction effect, and no significant difference was found in either condition. However, in terms of the average number of correct responses, the largest increase in the average number of correct responses from the pre-assignment and the highest average number of correct responses in the post-assignment were found in the voice condition. Therefore, it is suggested that the presentation of voice-based information has a greater effect on the retention of Chinese characters than the presentation of visual information.

Because this experiment was conducted with a fixed number of questions for each instructional condition, it is possible that differences in the difficulty of the questions may have influenced the difference in the range of performance improvements. However, if we consider the results of Q4 and Q5, it is considered that the participants tended to put more effort into recalling characters in the voice condition, which may have influenced the increase in the average number of correct answers in the post-guided questions. Therefore, hypothesis H4 was supported.

5 DISCUSSION

From the experimental results, it is found that the presentation of information by an interactive writing device with visual information and voice assistance provides a sense of comprehension and recall. Particularly, when the information is presented by voice assistance, the device could support users' active recall of Chinese characters while providing a high sense of understanding and support for the instructions.

As mentioned earlier, there is a challenge in writing support for Chinese characters to realize a framework that combines the convenience of information retrieval for daily use and the retention of retrieved content. The proposed system is characterized by a simple usable format by incorporating a retrieval function in the writing

Table 2: Question items and results for the questionnaire (N = 18, **:p<.01, *:p<.05, +:p<.10)

ID	Question	VoiceCond.(Vo)		VisualCond.(Vi)		Combined Cond. (C)		Main effect	Multi Comparison p-value (holm-Test)		
		Mean	SD	Mean	SD	Mean	SD		Vo-Vi	Vi-C	Vo-C
Q1	Instruction from the system is understandable.	6.11	0.25	6.56	0.12	6.72	0.11	**			*
Q2	I felt that I was receiving support from the device.	5.83	0.25	6.33	0.24	6.33	0.25	**			
Q3	I feel familiar with the system	4.06	0.33	3.39	0.31	4.39	0.27	*		**	
Q4	I feel I accomplished this Chinese character writing task by myself.	4.00	0.32	2.94	0.30	2.78	0.29	*	*		*
Q5	I tried by myself to recall Chinese characters that I could not write.	4.83	0.23	4.06	0.37	3.83	0.36	+	+		+

Table 3: Changes in the average number of correct answers (N=18) in the pre- and post-surveys for the questions asked in this survey (by condition of the guidance method).

Condition	All (24 questions)		Voice (8 questions)		Visual (8 questions)		Combined (8 questions)	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Mean	9.56	17.2	3.11	6.11	3.28	5.28	3.17	5.83
SD	0.70	0.87	1.45	1.64	1.02	1.52	1.25	1.33
p-value (t-test)	*(p<.05)		**(p<.01)		**(p<.01)		**(p<.01)	

tool, and then including the mechanism to enhance retention. The proposed framework for support via dialogue could be used as a means of supporting tasks requiring assistance, such as a person without sufficient knowledge learning another language on his/her own.

The contribution of this study is to clarify the usefulness of guidance based on human-like “speech” from the writing instrument as a method to encourage users to remember characters they cannot recall. Our study provide findings the effects of differences in recall and effort across modalities in information retrieval in realistic situation. As suggested by the results of the experimental questionnaire, visual presentation of information about the correct Chinese character form is considered the optimal solution for easy-to-understand assistance. However, even when the information was provided only by voice assistance, they still felt they had received easy-to-understand support; in addition, the system was able to promote active recall of the shapes of Chinese characters. In other words, this suggests the system may be able to promote users’ active behavior by using the irrational means of information transfer of “talking” to the user. These results provide a methodology for designing interfaces that encourage users’ proactive participation and effort by taking advantage of the modality effect. In the field of HCI, design methods that intentionally incorporate irrational elements to afford user behavior are being considered [25, 26]. Our findings suggest that voice-based guidance could be used as a means of designing systems to encourage users’ behavior in natural way.

6 CONCLUSION

In this paper, we proposed an interactive handwriting device that communicates with the user through natural language to provide information on the shapes of Chinese characters that the user cannot recall through voice assistance and visual presentation to support users’ character recall. Experiments with the proposed writing device showed that it is possible to support users’ independent recall of Chinese characters while providing a high level of understanding and a sense of support, especially when the information is provided through voice assistance. The user survey in this study focused on the recall of Chinese characters by native speakers and we need to additional survey for non-native speakers for clarify the effect of learning support. In addition, this study focused on explaining structured Chinese characters and examined the effectiveness of explanation based on its structure through speech. In future research, effective prompt design for explaining complex (or simple) Chinese characters that are difficult to structure needs to be considered. Thus, the result of this study have limitations. However, our findings suggest the provision of information with voice assistance is useful for enhancing users’ active behavior during learning.

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